

CSI473 Project Design Report

Campus Lost and Found Management System

Use this editable template for both phases. Replace guidance in square brackets and remove it before submission.

Field	Complete
Assignment/phase	Phase 1
Approved project title	Campus Lost and Found Management System
Approval decision/date	Approved - 17 August 2026
Team number	12
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Repository URL	https://github.com/Shirley-Thiite/csi473-project
Submission tag	phase1-submission
Date	11 September 2026

1. Executive summary

The Campus Lost and Found Management System is an online tool which facilitates the management of reporting, tracking and recovery of lost property on campus. Manual processes currently in place involve physically dropping off items, social media postings and word of mouth but are riddled with confusion, delay and have low success rate (approximately 28%). There is no centralized point of information, resulting in many items being lost forever.

The design of this system is described in this phase 1 report using the functional requirements (FR-01 through FR-29), business rules (BR-01 through BR-09) and quality scenarios (QS-01 through QS-03). The RBAC model combined with an unbreakable chain of custody has been implemented to ensure that the system complies with the law. Sequence and state machine modeling has been performed for UC-04: Report Item as Found.

Important design choices are manual inspection of claims to prevent fraudulent claims, non-reversible logging for integrity, and separate storage of private evidence for compliance reasons. The system will be integrated into existing authentication systems of universities with role-specific user interfaces for students, staff, security personnel, and administration. The vertical slice of reporting the lost laptop and retrieving it once found is completely designed and validated.

2. Problem evidence, stakeholders and scope

2.1 Problem Statement

Today, there are numerous cases when reporting lost items in university campuses happens in an unprofessional and inefficient way, including using physical locations or social media postings to report or find items. In such a way, there is no reliable source of information where people can post about their findings and losses. It leads to a problem of many items becoming unclaimed because of being unreported and forgotten. This project will create a unified system of reporting lost and found items online.

2.2 Evidence and Sources

Evidence/Source	Date/Origin	What It Shows
Observations of physical lost and found drop-off locations on campus	7 August 2026 (Library), 7 August 2026 (Student Center)	Physical lost and found boxes in buildings are often overflowing and disorganized, making it difficult to search for items efficiently
Evidence from campus social media groups (WhatsApp and Facebook)	Ongoing campus observations from 7 August 2026 to 31 August 2026	Students frequently post lost items on WhatsApp and Facebook groups, highlighting the lack of a centralized system for lost and found on campus
University Security Office records	2025 academic year	847 lost items reported; only 312 recovered (36.8% recovery rate)
Student Survey	2026	78% of students who lost items did not report them formally due to process complexity

Reconciliation of recovery rates: The 36.8% value only accounts for the items officially logged at the Security Office. However, the 28% value quoted in the executive summary is just an approximation of the recovery rate for all lost items, including those which never get officially logged. Both figures are therefore kept since they apply to two different groups: logged items vs. all lost items.

2.3 Approval Conditions

- Integration with university authentication system
- Adherence to university data protection guidelines
- Secure storage of personal evidence (photos, proof of ownership)
- Retention of audit trail for at least 3 years
- Mobile responsive design interface

2.4 Goals and Objectives

Goal	Objective	Success Metric
Goal: Design a software system that significantly increases efficiency, transparency, and success rate of returning lost items to their owners by providing a structured, digital, and user-friendly platform.		
Increasing Recovery Rate	Improve the percentage of reported lost items successfully returned to their owners	From current estimated 28% (36.8% for formally logged items) to at least 60% within 6 months of deployment
Improve User Experience	Provide a clear, intuitive, and secure process for reporting, searching, and claiming items	User satisfaction score of at least 4 out of 5 on post-interaction surveys
Reduce Time to Recover	Enable real-time reporting, instant search, and automated notifications for potential matches	Decrease average time to reunite owner with lost item from 5-7 days to under 48 hours

2.5 Stakeholders

Stakeholder/Role	Need	Concern	Evidence/Access
Student, Staff, User	Report a lost item, search for a found item, submit a claim for an item, and receive updates	Ease of use, security of personal data, speed of response, assurance that the claim process is fair	Common campus experience—students frequently lose items (phones, laptops, ID cards, keys) and rely on fragmented methods
Campus Administrator	Manage the system, verify and approve claims, oversee the handover process, generate reports	System integrity, prevent fraudulent claims, ensuring a clear audit trail for all transactions	Without formal verification, anyone could falsely claim expensive items

Campus Security	Physically handle and store found items, coordinate with administrators, manage unclaimed items	A clear, standardized process for intake, storage, and disposition to avoid confusion	Current systems often lack clear procedures, leading to items being misplaced or disposed of improperly
Campus IT/Developers	Ensure the system is reliable, secure, and maintainable	System uptime, data security and privacy, scalability to handle campus-wide usage, cost-effectiveness	

2.6 Scope

In Scope

- Registration of lost and found objects by authenticated users
- A database of lost and found objects that can be searched
- Claiming procedure for users to claim their found objects
- An administration tool for managing the objects and accepting/rejecting the claims
- Tracking of the chain of custody for all objects
- E-mail/notifications for informing the status
- Arrangement of handovers and confirmation of handovers

Out of Scope

- Integrating with the CCTV/security camera system of the campus
- Automated verification from the university-wide database of students
- Development of a native mobile application (the initial scope of development is a web-based solution)
- Support across multiple campuses/universities

2.7 Assumptions and Constraints

Assumptions

Assumption	Description
Network Availability	A stable campus network is available for system access
Browser Access	Users will have access to a modern web browser
Administrative Ownership	A designated campus office will act as the administrative owner of the system and manage the physical item handover

Constraints

Constraint	Description
Team Capacity	Development is bounded by a small student team and the number of teaching weeks available, which limits the number of features that can be built to production quality
Infrastructure	The system must run on infrastructure the team can provide
Time	The system must be designed and delivered as a semester-sized vertical slice within the CSI473 teaching timetable, not a full production rollout

3. Glossary

Term	Definition
Actor	A user role interacting with the system (Student, Staff, Security, Admin)

Chain of Custody	An immutable audit log tracking all actions performed on an item from report to resolution
Claim	A formal request by a user to recover an item they believe is theirs
Claimant	The user who submits a claim for an item
Found Report	A report submitted by a user who has found an item on campus
Handover	The physical transfer of an item from the system to the rightful owner
Item	A physical object that has been reported lost or found
Lost Report	A report submitted by a user who has lost an item on campus
Private Evidence	Confidential proof of ownership (serial numbers, unique marks, photos, receipts)
Report	A formal record of a lost or found item in the system
Resolution	The final outcome of a claim (accepted/rejected)
Verifier	The administrator/security personnel who reviews and validates a claim

4. Functional requirements and use cases

4.1 Functional Requirements

User Authentication & Profile (FR-01 to FR-03)

ID	Requirement	Priority
FR-01	System shall support authentication via university credentials	High
FR-02	System shall create a user profile upon first authentication	High
FR-03	System shall assign roles (Student, Staff, Security, Admin) based on university directory	High

Item Reporting (FR-04 to FR-08)

ID	Requirement	Priority
FR-04	System shall allow authenticated users to report items as lost	High
FR-05	System shall allow authenticated users to report items as found	High
FR-06	System shall require category, description, location, and date/time for all reports	High
FR-07	System shall allow users to attach private evidence to reports	Medium
FR-08	System shall prevent duplicate report submissions	High

Search & Matching (FR-09 to FR-12)

ID	Requirement	Priority
FR-09	System shall provide search by category, location, date, keywords	High
FR-10	System shall automatically suggest potential matches for administrators	Medium
FR-11	System shall display search results with relevant details	High
FR-12	System shall support pagination for large result sets	Low

Claiming & Verifying (FR-13 to FR-20)

ID	Requirement	Priority
FR-13	System shall allow users to submit claims on found items	High
FR-14	System shall record claim submission timestamp	High
FR-15	System shall require claimant to provide private evidence of ownership	High
FR-16	System shall require manual review of claims by administrators	High
FR-17	System shall notify users of claim status changes	High
FR-18	System shall allow one active claim per item per user	High
FR-19	System shall allow users to withdraw claims	Medium
FR-20	System shall record all claim decisions and reasons	High

Administration & Chain of Custody (FR-21 to FR-24)

ID	Requirement	Priority
FR-21	System shall record all item status changes in immutable log	High
FR-22	System shall require reason for all claim decisions	High
FR-23	System shall provide admin dashboard for oversight	Medium
FR-24	System shall support generation of audit reports	Medium

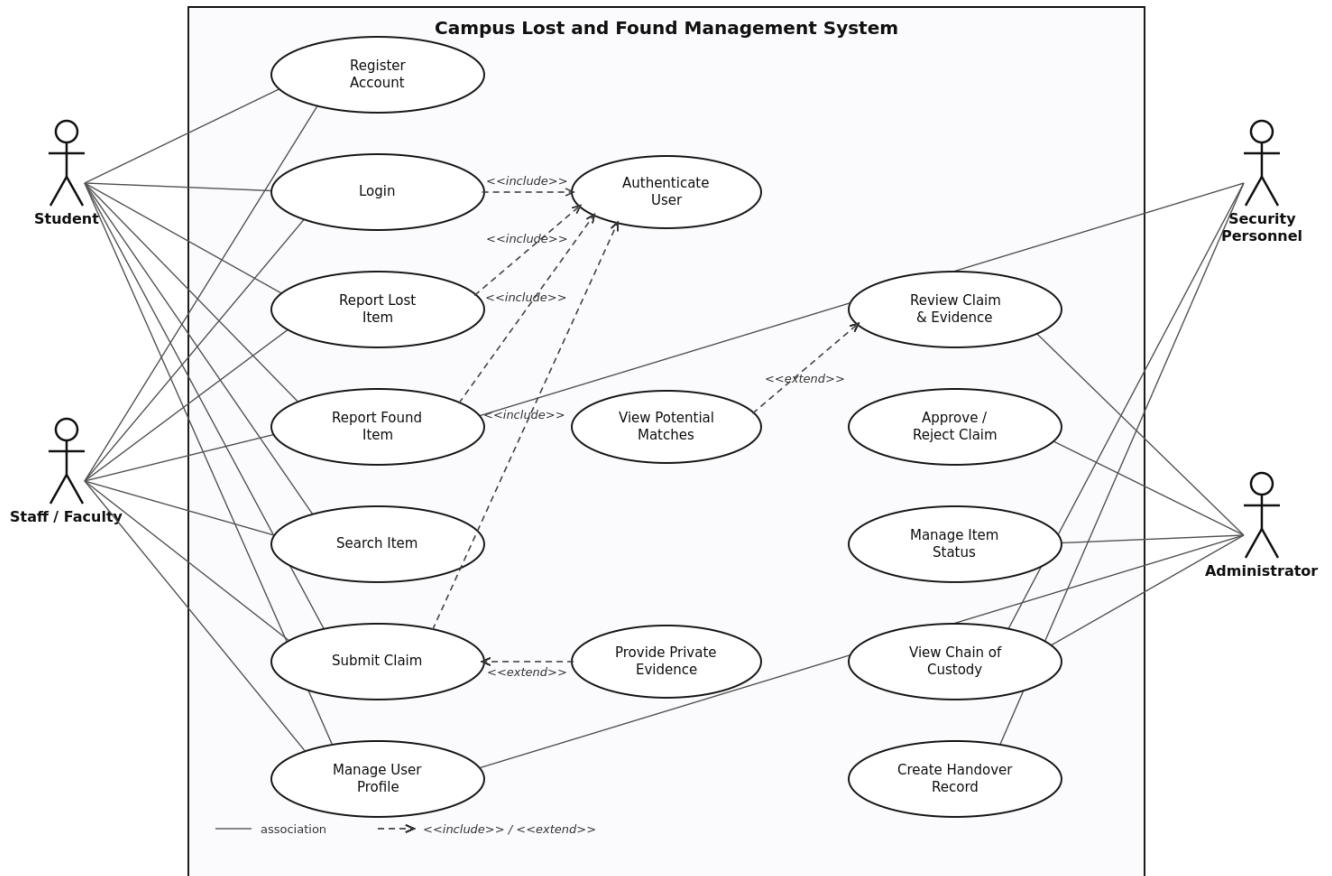
Security & Data Management (FR-25 to FR-28)

ID	Requirement	Priority
FR-25	System shall encrypt all private evidence at rest	High
FR-26	System shall encrypt all data in transit (HTTPS)	High
FR-27	System shall enforce role-based access control	High
FR-28	System shall retain audit logs for minimum 3 years	High
FR-29	System shall provide a mobile responsive interface	High

4.2 Actors

Actor	Description
Student	Current university student with university email
Staff/Faculty	University employee with university email
Security Personnel	Campus security staff responsible for found items and verification
Administrator	System oversight, claim verification, and reporting

4.3 Use Case Diagram



4.4 Detailed Use Case: UC-04 Report Item as Found

Attribute	Description
ID	UC-04
Name	Report Item as Found
Primary Actor	Security Personnel, Authenticated User
Secondary Actors	System
Precondition	User is authenticated via university credentials
Postcondition	Found report is created with FOUND status; custody log updated
Trigger	A user finds an item on campus and wants to log it in the system

Basic Flow

Step	Actor Action	System Response
1	User navigates to "Report Found Item" page	System displays found report form
2	User enters item details (category, description, location, date/time)	System validates input fields
3	User uploads private evidence (optional)	System validates file type and size
4	User submits form	System validates all fields; prevents duplicates
5	-	System creates FoundReport with unique ID
6	-	System updates Item status to FOUND
7	-	System creates ChainOfCustody entry: FOUND
8	-	System sends success notification to user
9	System displays confirmation page	-

Alternative Flows

Flow	Description
A1: Duplicate Report	System detects a potential duplicate report. User is notified and asked to review existing reports before submitting.
A2: Invalid Data	System validates field formats (e.g., invalid date, category mismatch). User is prompted to correct data.
A3: System Failure	System fails to save report. User receives error notification and is instructed to contact support.
A4: Evidence Upload Failure	File exceeds size limit or is unsupported format. User is notified and can retry.

4.5 Acceptance Criteria

Criterion	Verification Method
AC-04-01: User must be authenticated to report a found item	Test: Attempt report without login → rejected
AC-04-02: All required fields must be completed	Test: Submit empty form → validation errors
AC-04-03: Category must be from valid list	Test: Submit invalid category → rejected
AC-04-04: Duplicate reports are detected	Test: Submit same item twice → duplicate warning
AC-04-05: Report receives unique identifier	Test: Verify ID format after submission
AC-04-06: Chain of custody entry is created	Test: Check log after submission → entry exists
AC-04-07: Private evidence is stored securely	Test: Verify encryption for uploaded files

5. Quality requirements

5.1 Quality Scenarios

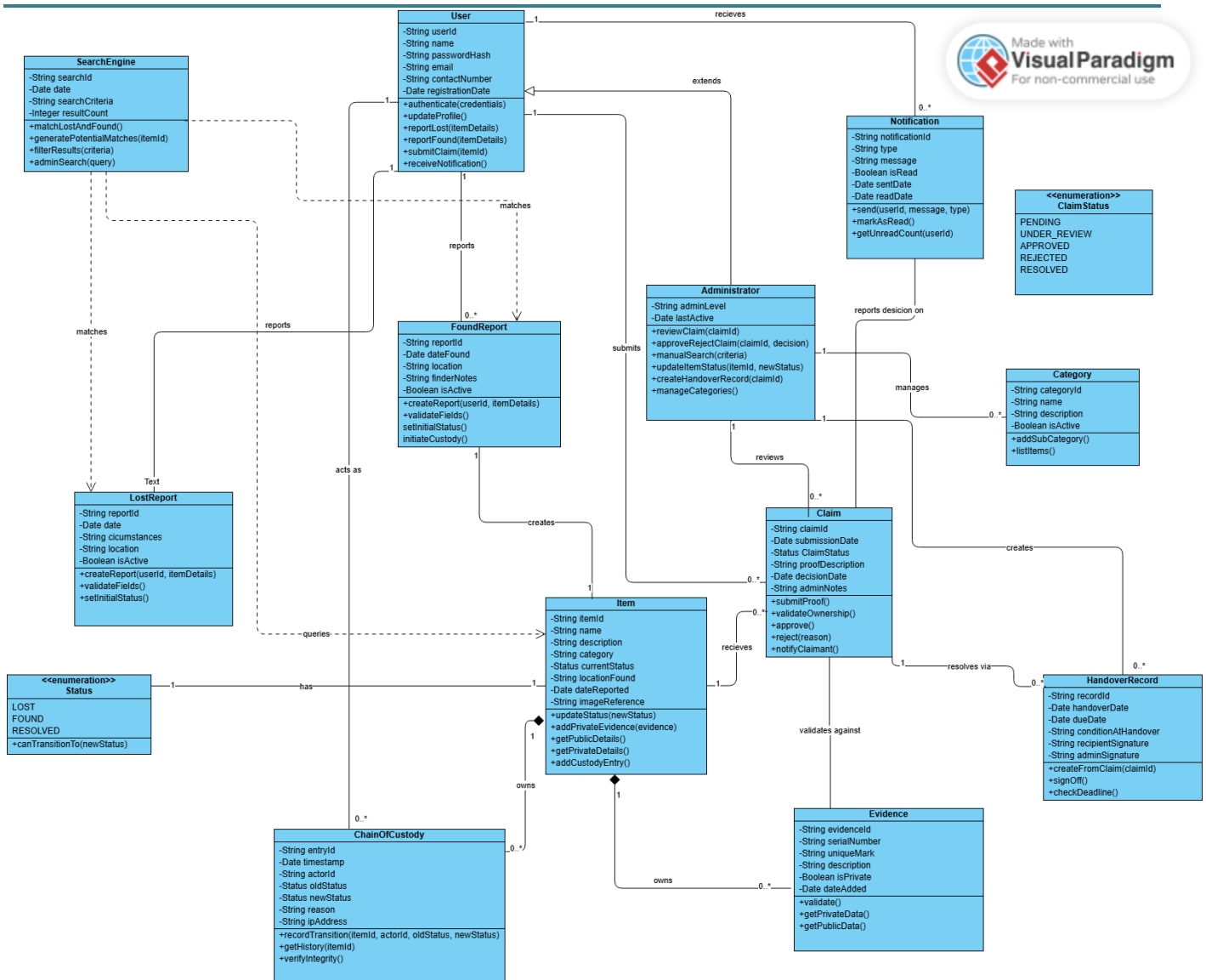
Quality Attribute	Target	Verification Method	Design Obligation
Performance	Search ≤ 3 seconds for 10,000 items	Load testing with JMeter	Indexed search, pagination
Availability	$\geq 99\%$ uptime (8AM-10PM daily)	Monitoring	Redundant hosting, health checks
Security	100% private evidence encrypted at rest	Security audit	AES-256 encryption
Usability	$\geq 90\%$ success rate for report creation	User testing (n=20)	Intuitive UI, clear validation
Maintainability	Category addition ≤ 30 minutes	Developer testing	Admin panel with category management
Reliability	$\geq 99.9\%$ error recovery	Fault injection	Transaction rollback, error logging
Data Integrity	0% data loss for committed transactions	Audit review	ACID compliance, regular backups
Accessibility	WCAG 2.1 AA compliance	Automated + manual testing	Screen reader support, color contrast
Transparency	All custody actions visible to admins	Audit review	Immutable Chain of Custody logs
Privacy	Private evidence is visible only to administrators	Security audit	Role-based access + encryption

5.2 Measurable Metrics

Metric	Target	Measurement Method
Report creation time	≤ 3 minutes	User testing timing
Claim processing time	≤ 48 hours	System tracking
Search result accuracy	$\geq 85\%$ relevant results	User feedback survey
System response time (page load)	≤ 2 seconds	Performance monitoring
User satisfaction	$\geq 4.0/5.0$	Post-task survey
Mobile responsiveness	100% of features on mobile	Device testing matrix
Recovery rate improvement	$\geq 60\%$ within 6 months	System tracking

6. Analysis/domain model and responsibilities

6.1 Domain Model



6.2 Business Rules

ID	Rule	Implementation
BR-01	Reporting uniqueness prevents duplicates	Duplicate detection algorithm on report submission
BR-02	Status transition order: LOST → FOUND → CLAIMED → VERIFIED → HANDOVER_PENDING → RESOLVED	State machine enforcement
BR-03	Users cannot claim their ~~reported~~ own reported items	Validation on claim submission
BR-04	Private evidence protected from unauthorized use	Encryption + RBAC
BR-05	Handover within 7 days after approval	Scheduled job + reminders
BR-06	Authentication needed for all protected actions	Middleware checks
BR-07	Completeness of chain of custody guarantees audit trail	Immutable logging
BR-08	Only one active claim per item per user	Database constraint + validation
BR-09	Automated matching provides suggestions only; no automatic claims	Manual admin review required

6.3 CRC Cards

CRC Card 1: Item

Aspect	Details
Class Name	Item
Responsibilities	• Track its own status and location • Manage private evidence • Update status when validated • Maintain chain of custody entries
Collaborators	• Status • Evidence • ChainOfCustody • LostReport • FoundReport
Rationale	Item is the central entity and should own its lifecycle data to maintain cohesion

CRC Card 2: Claim

Aspect	Details
Class Name	Claim
Responsibilities	• Manage claim lifecycle (submission, approval, rejection) • Validate ownership proof • Handle status transitions • Notify claimant of decisions
Collaborators	• Item • User • Evidence • HandoverRecord • Notification
Rationale	Claim owns its approval/rejection process to maintain data integrity

CRC Card 3: ChainOfCustody

Aspect	Details
Class Name	ChainOfCustody
Responsibilities	• Record immutable audit trail of all status changes • Maintain actor identity and timestamps • Provide complete history for an item • Ensure tamper-proof records
Collaborators	• Item • User • Status
Rationale	Separate class for auditing concerns maintains separation of concerns and supports immutability

CRC Card 4: User

Aspect	Details
Class Name	User
Responsibilities	• Authenticate using credentials • Manage profile information • Report items lost or found • Submit claims for found items • Receive notifications
Collaborators	• LostReport • FoundReport • Claim • Notification • Role
Rationale	User is the primary actor and owns its actions and interactions with the system

CRC Card 5: FoundReport

Aspect	Details
Class Name	FoundReport
Responsibilities	• Create new found item record • Validate required fields • Set initial status to FOUND • Initiate chain of custody entry • Generate unique report ID
Collaborators	• Item • ChainOfCustody • User (reporter)
Rationale	FoundReport specializes the reporting process with specific found item workflows

CRC Card 6: Administrator

Aspect	Details
Class Name	Administrator (extends User)
Responsibilities	• Review and approve/reject claims • Manually search database • Update item statuses • Create handover records • Manage system configuration (categories)
Collaborators	• Claim • Item • HandoverRecord • ChainOfCustody • Category
Rationale	Administrator role has elevated privileges and specific responsibilities that extend base User capabilities

CRC Card 7: Notification

Aspect	Details
Class Name	Notification
Responsibilities	• Send alerts for potential matches • Notify claim decisions • Track read/unread status • Manage notification types
Collaborators	• User • Claim • Item
Rationale	Notification is responsible for user communication and should be decoupled from business logic

CRC Card 8: SearchEngine

Aspect	Details
Class Name	SearchEngine
Responsibilities	• Match lost and found items • Generate potential matches • Filter search results • Support administrative manual search
Collaborators	• Item • LostReport • FoundReport • Administrator
Rationale	Search functionality is separate from item management to support scalability and maintainability

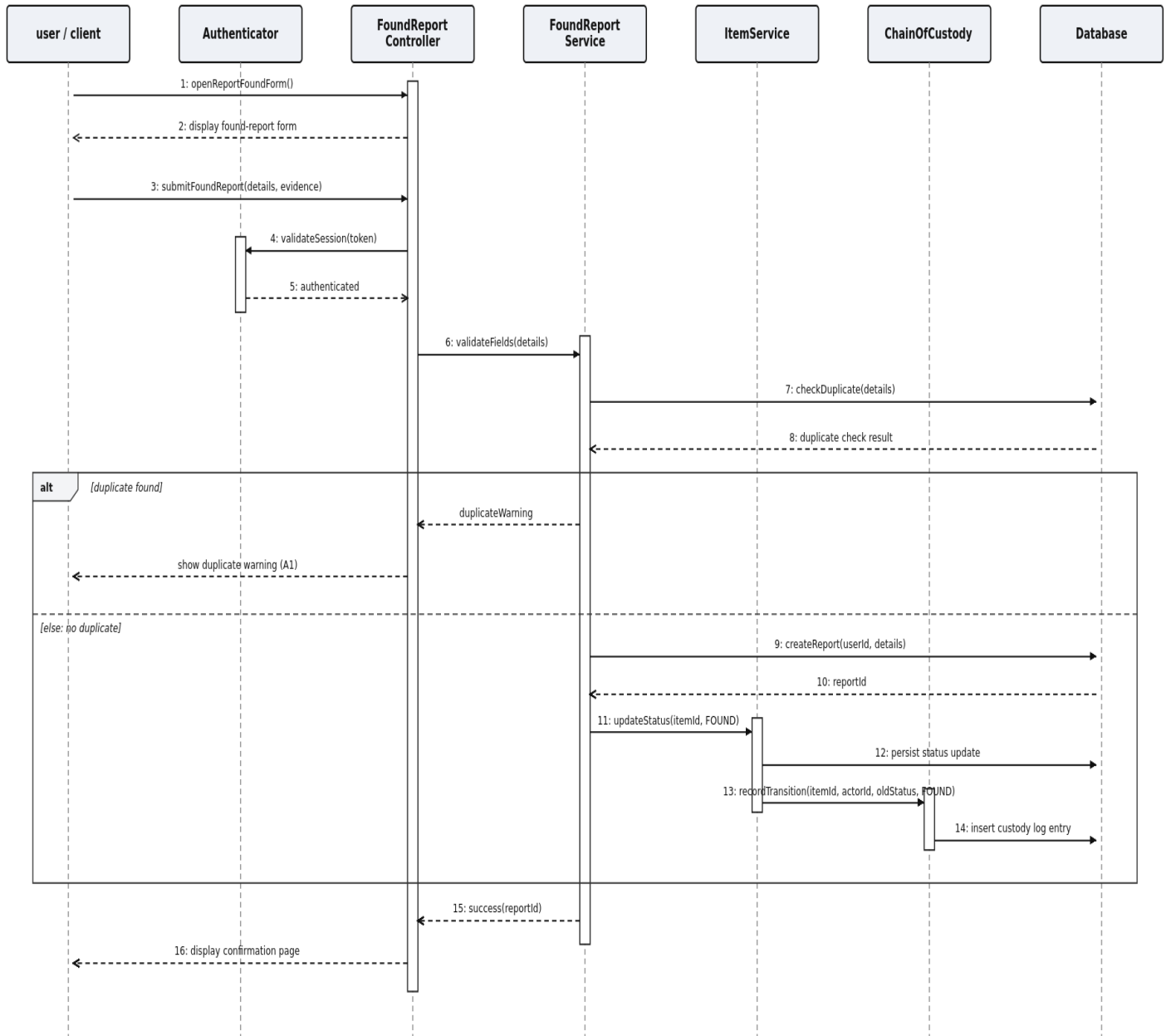
Responsibility Allocation Rationale

The responsibility allocation rationale complies with the principles of domain-driven design:

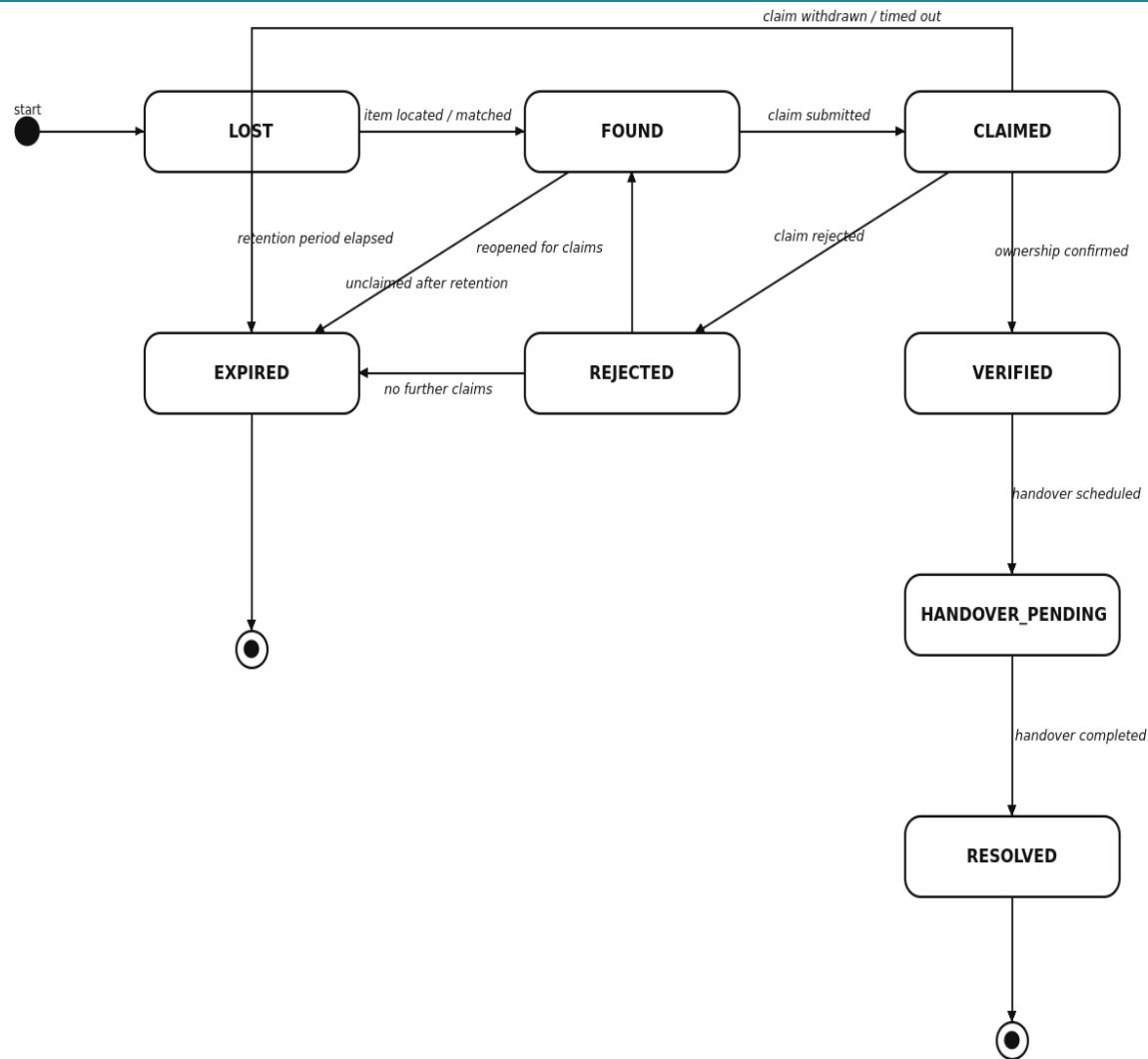
- Item is the aggregate root and controls all state changes
- ChainOfCustody is implemented as an append-only log to ensure immutability
- Claim controls the verification process to separate the review logic from item management
- Evidence handling is delegated to different report types to separate concerns appropriately
- Administrator and Security have different responsibilities to avoid conflict of interest

7. Interaction and lifecycle

7.1 Sequence Diagram: UC-04 Report Item as Found



7.2 State Machine Model: Item Lifecycle



8. Architecture and decisions

8.1 Architecture Drivers

#	Driver	Source	Quality Attribute	Constraint
AD-01	Auditability and non-repudiation — Every item status change must be recorded in an immutable, tamper-evident log retained ≥ 3 years	FR-21, FR-28, BR-07, Approval condition 4	Integrity, Compliance	University data protection; legal retention
AD-02	Secure handling of private evidence — Ownership proof (photos, receipts) must be encrypted at rest and in transit, accessible only to authorised roles	FR-15, FR-25, FR-26, BR-04, Approval conditions 2 and 3	Security, Privacy	University data protection guidelines
AD-03	Integration with university identity and mobile access — Authentication must use university credentials, and the interface must be mobile-responsive	FR-01, FR-03, FR-29, Approval conditions 1 and 5	Interoperability, Usability	Existing university auth system; mobile approval condition

8.2 Architecture Alternatives

Alternative	Description	AD-01 Auditability	AD-02 Evidence Security	AD-03 Integration/Mobile	Consequence Accepted

A: Modular Monolith with Event-Sourced Audit Log	Single deployable web app; audit log implemented as append-only event store; RBAC middleware; server-rendered responsive views + REST API for future mobile	Strong — event sourcing natively immutable	Strong — encrypted blob store for evidence	University SSO via OIDC; responsive CSS framework	Simpler deployment for small team; may need extraction of audit service if scale grows
B: Microservices with Separate Audit and Evidence Services	Independent services: Auth, Item, Claim, Audit, Evidence; each with own datastore; API gateway; SPA front-end	Strong — dedicated audit service with write-once storage	Strong — evidence service with KMS-backed encryption	SSO service; SPA mobile-responsive	Higher operational complexity than team can sustain in one semester; over-engineering risk

Phase 2 Recommendation: Alternative A – A modular monolith with an event sourced audit log fits both small team and semester sized vertical slice constraints, while meeting all three architectural forces. Dedicated audit/evidence services in Alternative B could be extracted if required at a later stage.

8.3 Design Rationale Record

Decision	Chosen	Alternative Considered	Consequence Accepted
RBAC design	Role-based access control with 4 roles	Attribute-based access control (ABAC)	Simpler to implement and audit; less granular than ABAC
Immutable chain of custody	Append-only log with no update/delete	Mutable log with version history	Stronger integrity guarantee; higher storage growth
Manual claim inspection	Admin reviews every claim	Automated matching with auto-approval	Fraud prevention; higher admin workload
Separate private evidence storage	Dedicated encrypted blob store	Evidence stored inline with claim record	Better security isolation; added architectural complexity
University auth integration	OIDC SSO with university directory	Local account system	No password management burden; dependency on external system
Automated matching	Suggestions only, no auto-claims	Auto-approve high-confidence matches	Prevents fraudulent auto-claims; slower resolution
State-machine enforcement	Explicit FSM in Item (BR-02)	Implicit status flags validated ad hoc	Stronger invariant protection; must update FSM for every new state

9. Submission Evidence

Evidence	Location
Final Phase 1 PDF	submissions/CSI473_A2_Phase1_Team12.pdf
Phase 1 review checklist	docs/phase1-review-checklist.md
Architecture drivers note	docs/architecture-drivers.md
Editable model sources	docs/*.drawio, docs/*.puml
Readable exports	docs/*.svg
Repository tag	Phase1-submission